



Allerton
has them
now!

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Q1

~Code~
340



when was the
last time you
saw a hummingbird?



CS 340

Building Blocks 0b01



Updates

1. MP 1.1, MP 1.2, MP 1.3
 - a. Switch to address sanitizer

MP 1.2 ↗

2. Environment check-off
 - a. Check your PL grade

3. HW 2 is out → wed

4. Collaboration time/office hours today

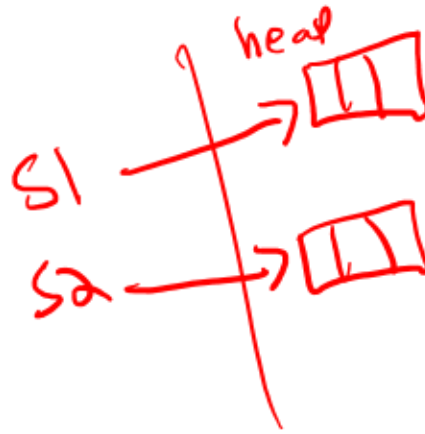
→ next week tuesday

1. you check a box
wait
2. upload grades
3. you can see grade

3:20 - 4:00

Review. What prints?

```
3  int main(){
4      char *s1 = malloc(3);
5      s1[0] = 'C';
6      s1[1] = 'S';
7      s1[2] = 0;
8      char *s2 = malloc(3);
9      s2[0] = 'C';
10     s2[1] = 'S';
11     s2[2] = 0;
12     if (s1 == s2){
13         printf("yay");
14     }
15 }
```



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Q2

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are the
pointers holding
the same address

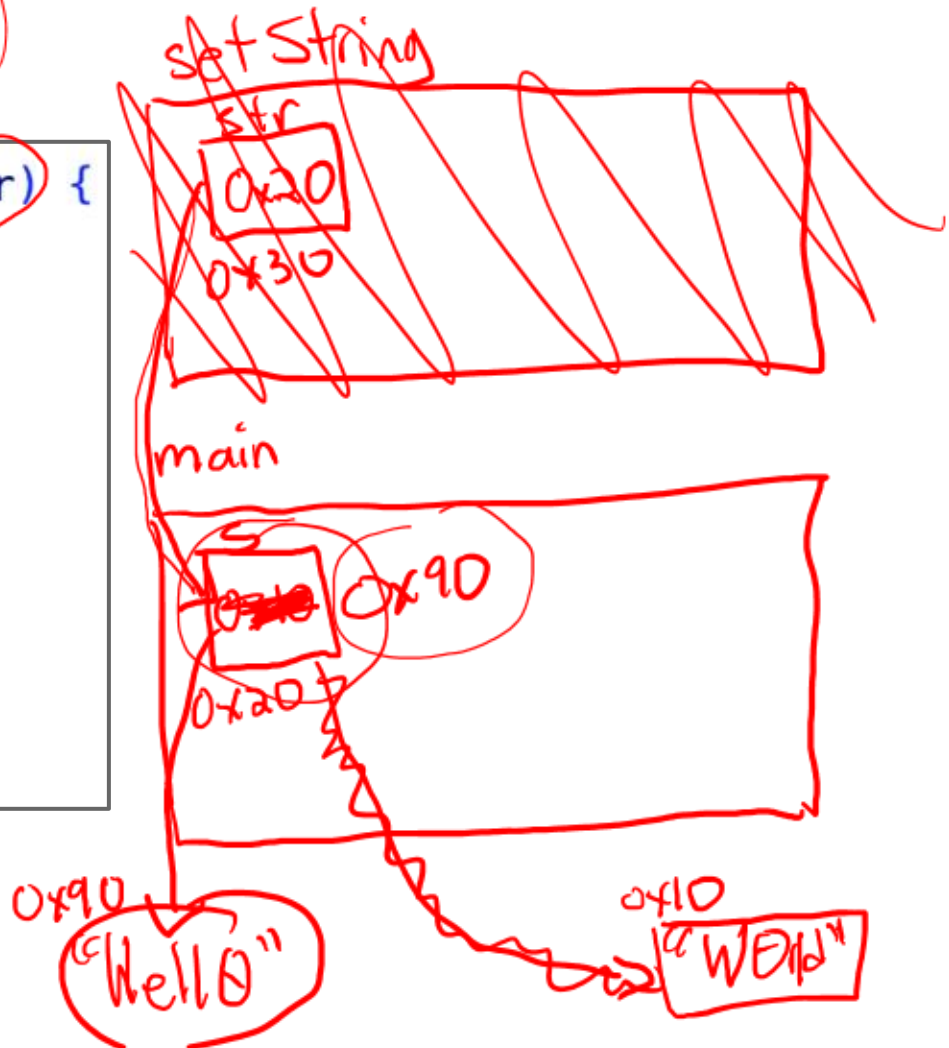
nothing

Review

$\text{char } '0' \neq 0$

```
22 void setString(char **str) {
23     *str = "Hello";
24 }
25
26 int main() {
27     char *s = "World";
28     setString(&s);
29     printf("%s", s);
30     return 0;
31 }
```

Hello



Building Blocks Ob01

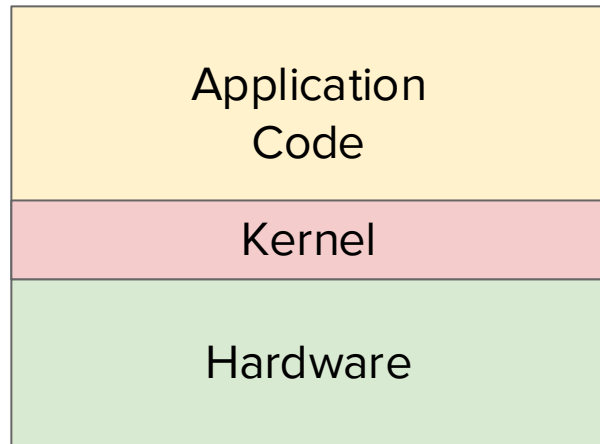
LGs:

- Be able to interpret basic gate diagrams
- Be able to convert between decimal and binary

Mental Models Developed

How a computer works at a physical level

Why and how binary can be used to store information



Agenda



1. Circuit and gate basics
2. Binary

not a
physics
class

Circuit and Gate Basics

Idea 1: Nodes Hold Voltage

Node - example $\rightarrow$ wire, end of a battery

Voltage - a state - electric potential energy.

Current - movement - charge moving $\rightarrow$ equalize voltage



Idea 2: Two Important Nodes

Power - (V) — battery side 1

Ground - (G) — battery side 2

Don't directly
connect

Idea 3: Transistors are voltage-controlled switches



PMOS
dot... default down



NMOS
default up



Big Ideas

Nodes hold voltage.

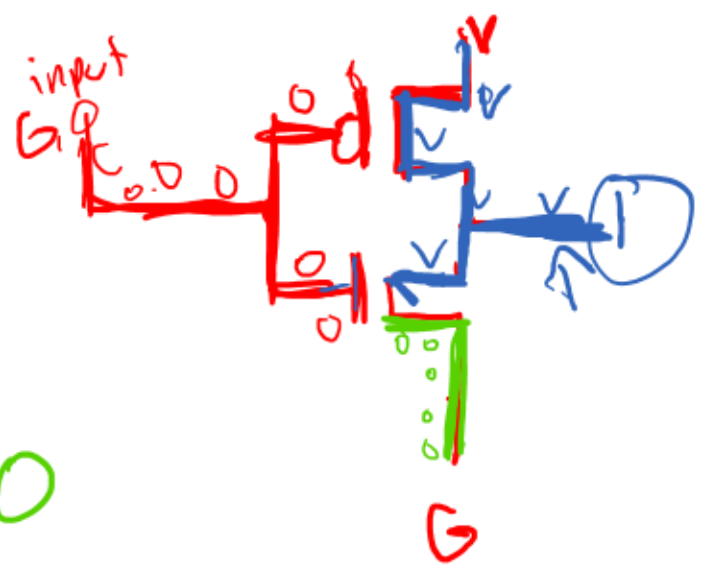
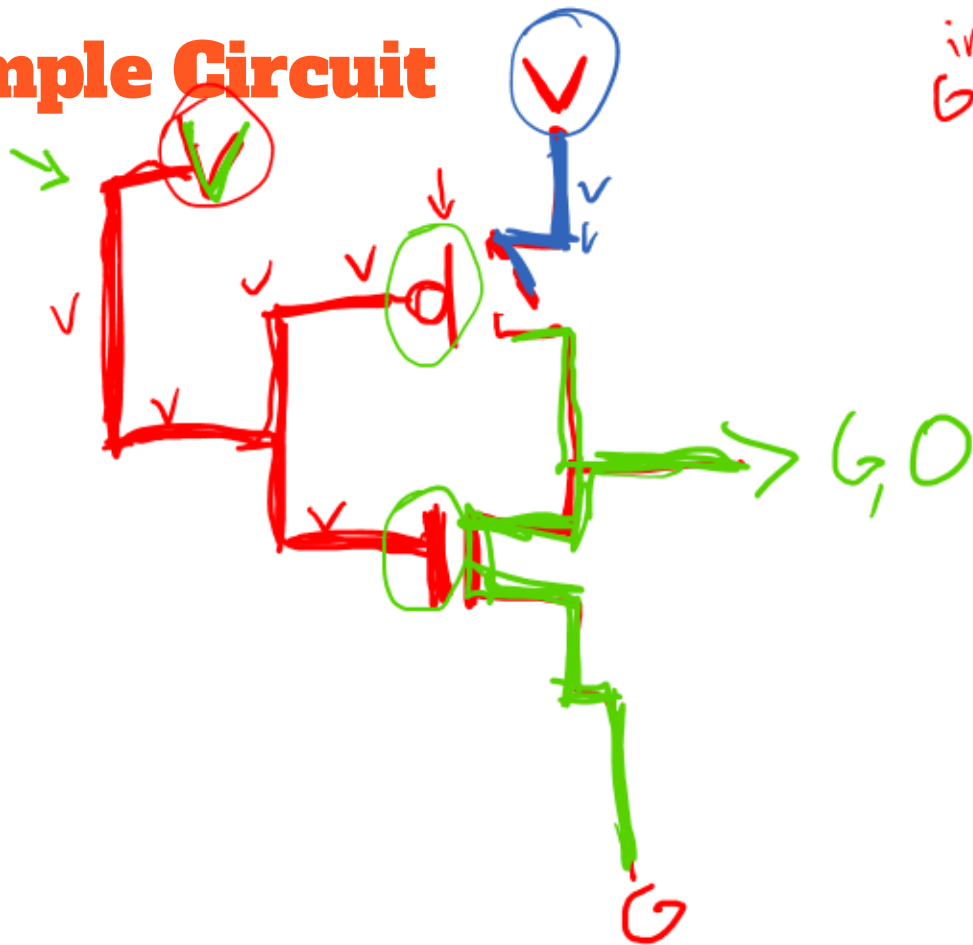
↪ have a state
high or low
✓

Current flows from high to low voltage to equalize voltage.

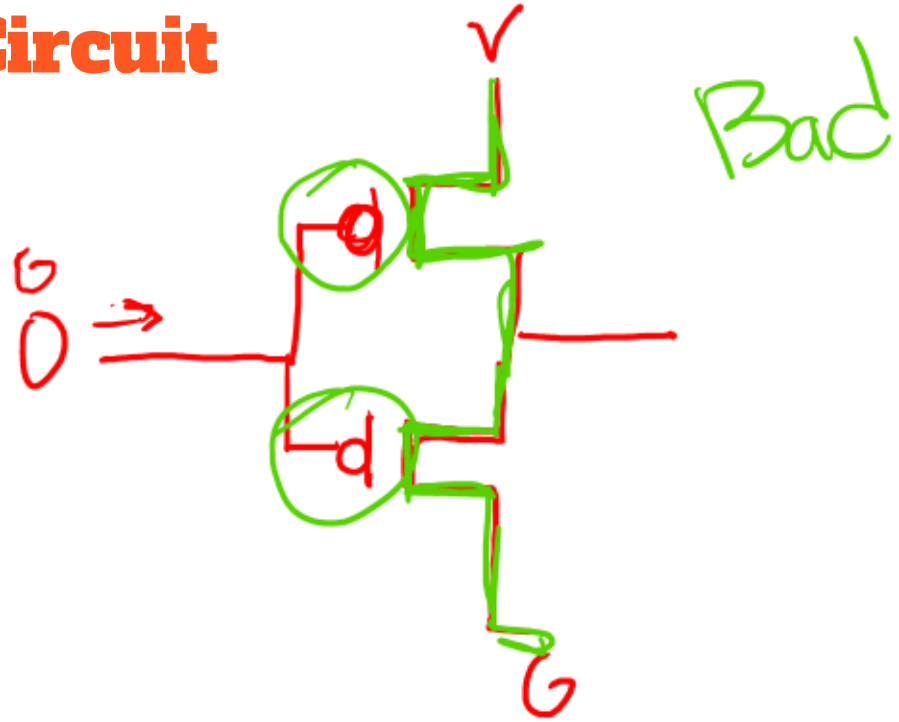
Power and ground are two important nodes needed.

Transistors are voltage-controlled switches.

Example Circuit



Short Circuit



What voltage does the node C hold?

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Q3

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Gates

Use input voltage states to change the... transistors
and then the output
state

Gates use... transistors to control output voltage

Gates can be used to... implement logic
AND, OR, NAND, NOR, XOR

Logic Review

XOR

A	B	XOR
0	0	0
1	0	1
1	1	0
0	1	1

A	B	AND
0	0	0
1	0	0
1	1	1
0	1	0

A	B	NOR
0	0	1
1	0	0
1	1	0
0	1	0

[illegible]

A

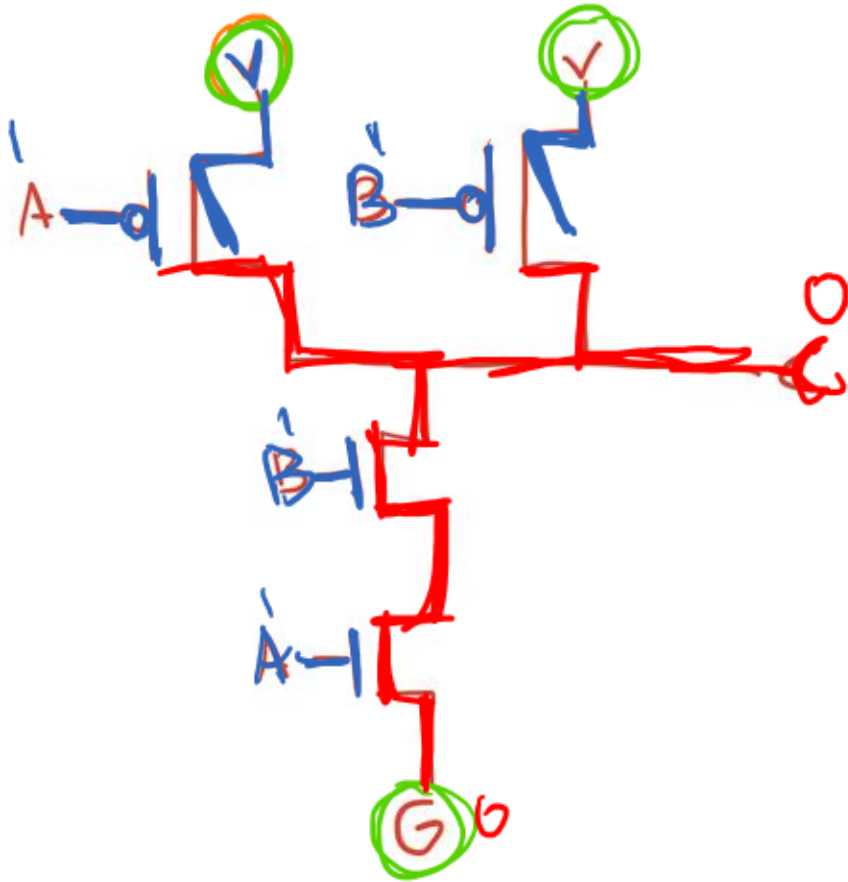
if A is
C = 0
D = 1

G.O

66

66

Example Gate



input is A, B
output is C

A is 1

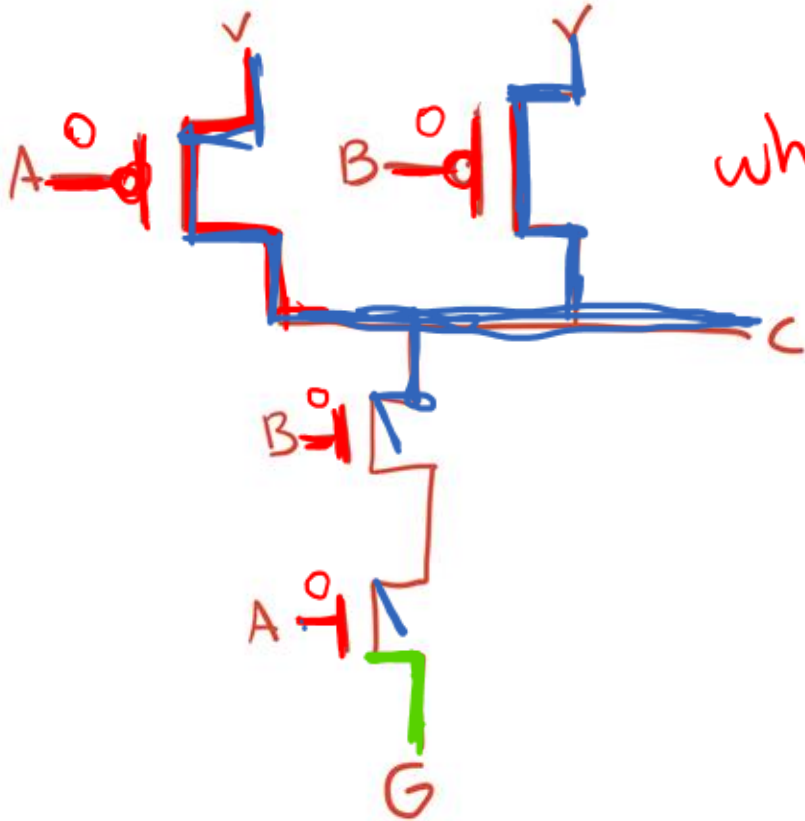
B is 1

what is C?

C is 0
when A and B are 1

Example Gate

if $A == 0$
 $B == 0$
what is C ?



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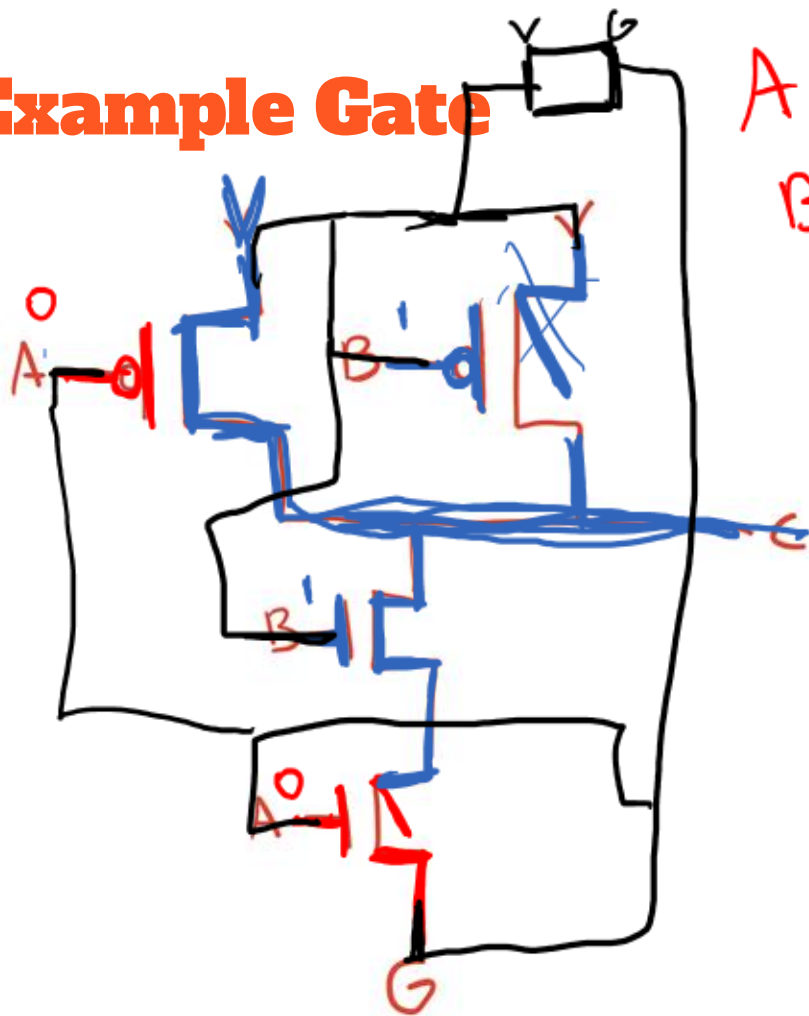
Q4

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C is 1

Example Gate



A is 0
B is 1 ?

C = 1

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Q5

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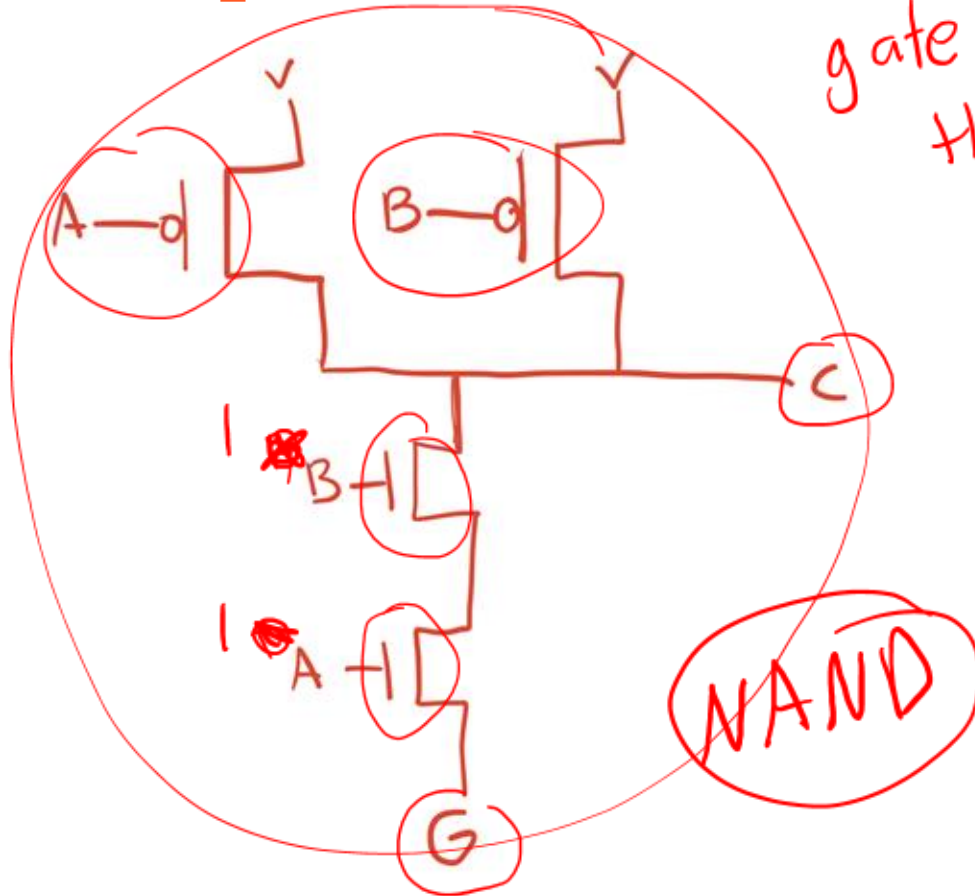
Example Gate

what logic
gate is
this?

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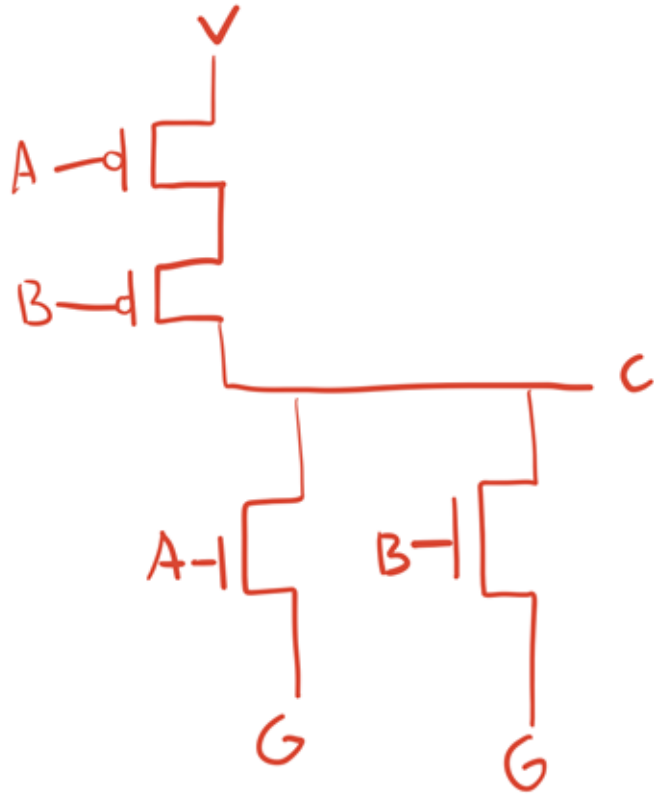
Q6

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A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

Example Gate



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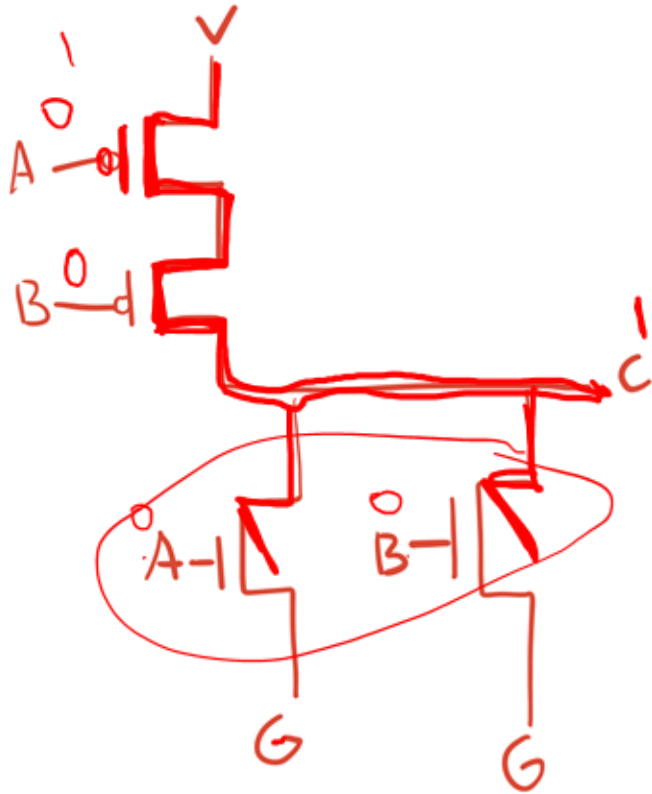
Q7

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Example Gate

What logic gate
is this?



A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

NOR

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Q8

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Big Picture

NAND

electricity $\rightarrow$ logic gates $\rightarrow$ math, selection, etc...

We can make any truth table logic

$\overset{1}{V}, \overset{0}{6} \rightarrow 1, 0 \rightarrow$ information is stored
as high or low Voltage

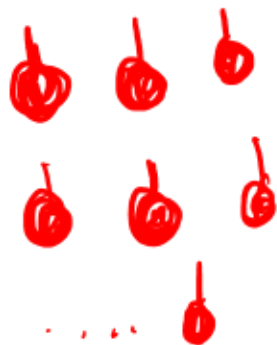
$\Rightarrow 1, 0$ binary number system

The Binary Number System

Number Systems

infinite amounts
finite digits

0, 1, 2, 3, 4, 5, 6, 7, 8, 9



← 5

3	2	1	0
		1	0
		10^1	10^0
		10	0
+			
= 10			

Base-2 Number System

0, 1

1

1

1 1

1 0
↑ ↑
 2^1 2^0

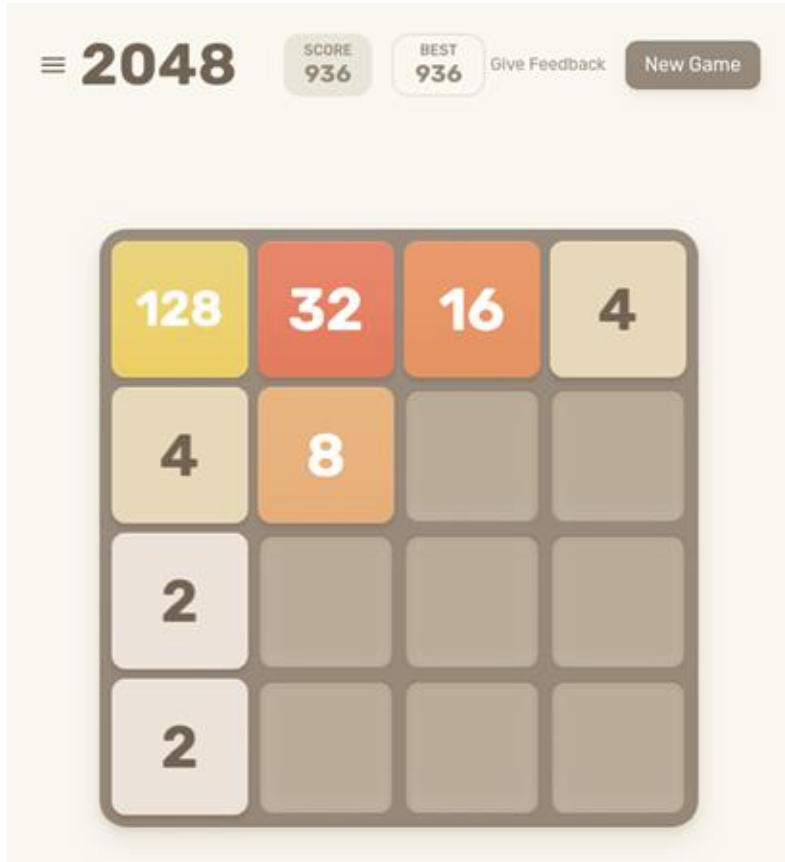
$$2 + 0 = \textcircled{2}$$

Going from Binary to Decimal

0	1	0	0	1	0	1
32	16	8	4	2	1	
			↑			↑

$$32 + 4 + 1 = 37$$

Powers of 2 Practice....



What is 0b10111 in Decimal?



$$16 + 4 + 2 + 1 = 23$$

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Going from Decimal to Binary

67

→

0	1	0	0	0	0	1	1
128	64	32	16	8	4	2	1

00067

$$\begin{array}{r} 67 \\ - 64 \\ \hline 3 \\ - 2 \\ \hline 1 \end{array}$$

↓
0b1000011
0b0000100011

What is 14 in binary?

→ 0 0 0 1 1 1 0
64 32 16 8 4 2 1

14
- 8

6
- 4

2
- 2

0

0b1110

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Q10

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Counting in Binary

0000 0

0001 1

0010 2

0011 3

0100 4

0101 5

What is 0b0001 + 0b1001 in decimal?

0b1001
8 4 2 1

10

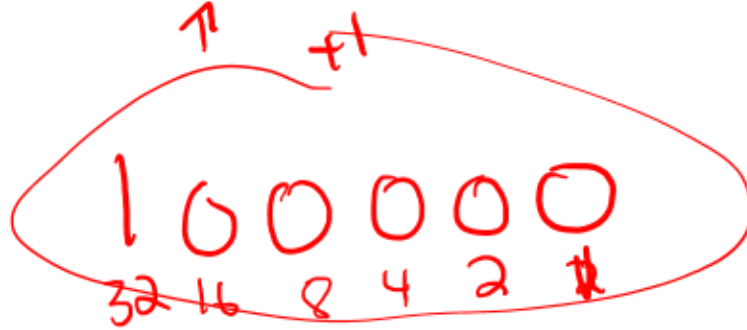
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What is $0b11111$ in Decimal?



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Q12

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31

Addition in Binary

$$\begin{array}{r} 1 \\ 560 \\ + 560 \\ \hline 1120 \end{array}$$

$$\begin{array}{r} 1 \quad 11 \\ 11011 \\ 11001 \\ \hline 110100 \end{array}$$

**Bits take up space in hardware.
How many bits of space would I
need to fully represent the
result of 0b11011 + 0b11001?**

~~~~~ 5 bits

need 6 bits

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char 010
char 110

Subtraction in Binary

Big Takeaways

What do these bits mean?

0b0100101

can't tell

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What is 100 in Decimal?

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Q15

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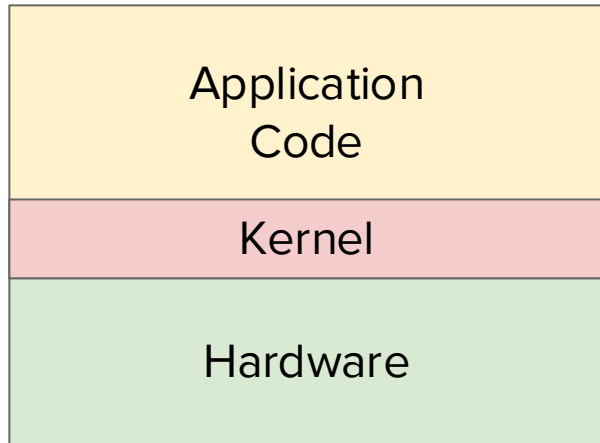
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